Findings showed decreased electrical resistance in colder periods but an increase in warmer seasons, suggesting improved battery health with rising temperatures. Automakers usually rely on traditional BMS algorithms developed in labs, where machine learning tracks the performance metrics of a single 4-volt battery cell, consistently charging and discharging at a fixed temperature until depletion. However, the data from Audi came from a 396-volt battery setup with 384 cells.

Dual-arm robot learns bimanual tasks from simulation

Scientists at the University of Bristol, based at the Bristol Robotics Laboratory, have designed the Bi-Touch system, which enables robots to perform manual tasks by receiving guidance from a digital assistant. Researchers developed a virtual simulation with two robot arms equipped with tactile sensors, utilising reward functions and a goal-update mechanism to encourage mastery of bimanual tasks. They then created a real world dual-arm robot system where the trained agent could be directly applied, acquiring skills through deep reinforcement learning (Deep-RL), a technique that facilitates learning via trial and error. In this setup, the robot learns decision-making by experimenting with various actions to complete tasks, such as handling delicate objects without damage. Relying solely on proprioceptive feedback, the AI agent proficiently learns to lift fragile items, even as delicate as a Pringle crisp, through iterative success and failure assessments.

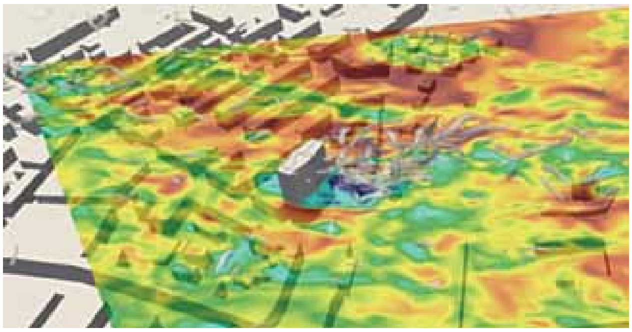
Dual-arm robot holding crisp (Credit: Yijiong Lin)

Advancing robot grips through deep learning

Researchers at the University of Bonn have unveiled a learning framework to enhance a robot arm's object manipulation skills for functional purposes. The refined pre-grasp manipulation method employs deep reinforcement learning, a proven effective strategy for training AI algorithms. Through this approach, the team trained a model to adeptly manage objects before firmly grasping them, ensuring the robot holds them as intended, using a nuanced dense reward function that promotes better finger-object engagement and alignment with the functional grasp specified. This methodology is believed to be adaptable across various robotic arms and hands, facilitating the handling of differently shaped objects and potentially being tested on numerous physical robots. Demonstrating that the replication of complex human-like actions is possible with a single computer and several hours of training, the team plans to transition this learned model into practical applications.

Advanced lidar-based turbulence model for wind testing

Vind-Vind company has introduced an innovative turbulence modelling technique that promises to reshape how large structures interact with natural wind conditions. The company is tackling the deficiencies of unidirectional wind tunnel testing by developing a turbulence model that leverages real-world data collected by a 10ns pulse lidar system. To manage the substantial data generated, they abandoned the initially considered FPGA solution for spectrum's SCAPP drivers, facilitating data transfer rates up to 12.8GB/s directly to a CUDA-based graphics card, bypassing the traditional CPU route. This setup, utilising an Nvidia Quadro A4000 graphics card with 6,144 cores, significantly outperforms standard CPUs. The initiative aims to scrutinise the similarity between actual and simulated urban environment turbulence. Plans are underway for further enhancements and data-backed validation.



Wind interacts with buildings in very complex ways. This screenshot from the video shows wind velocities at 33m height relative to the innovative Lidar system located at the upper left corner (Credit: https://spectrum-instrumentation.com)

Fujitsu develops advanced 5G millimetre-wave chip for radio units

Fujitsu Limited has announced a new millimetre-wave chip for 5G that can support multibeam multiplexing (sans polarisation multiplexing), allowing a single chip to multiplex up to four beams for the RU in 5G base stations. When applied to real base stations using the new technology, the company has shown the capability to attain communications speeds of 10Gbps or higher while occupying only half the space of